Towards an integrative typology of tone

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Where is tone in typology?

half or more of the world's languages make use of pitch to distinguish morphemes
(cf. Yip 2002)

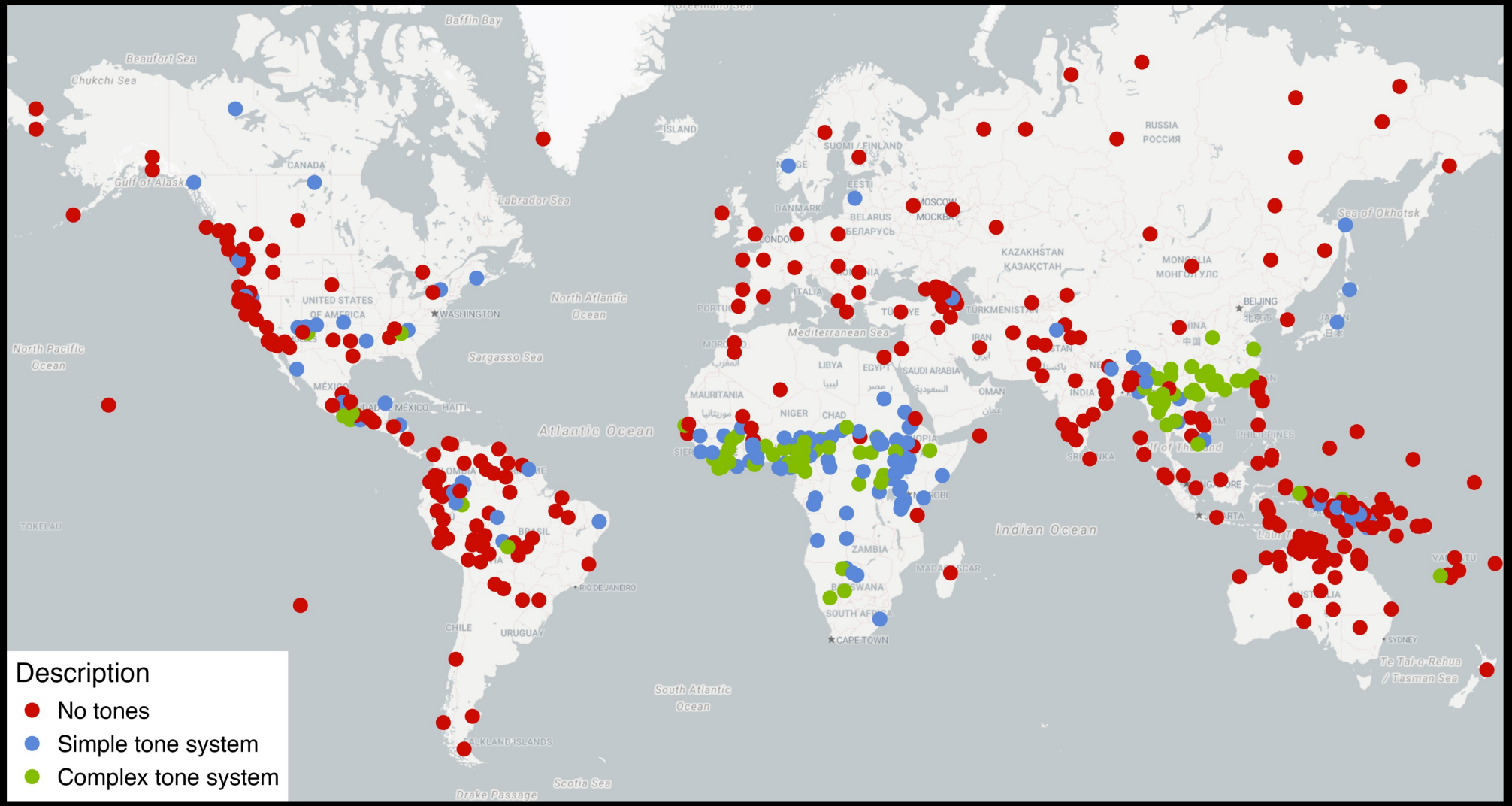
but tone is not well represented in typology

phonology is underrepresented in typology even though it was the starting point
(cf. Moran et al. 2023)

"no other phonological feature is treated with such indifference as tone"
(Hyman 2011)

Hyman 2011. *HB Phon Theory* | Moran et al. 2023. *Ling Typ 27(2)* | Yip 2002. *Tone*

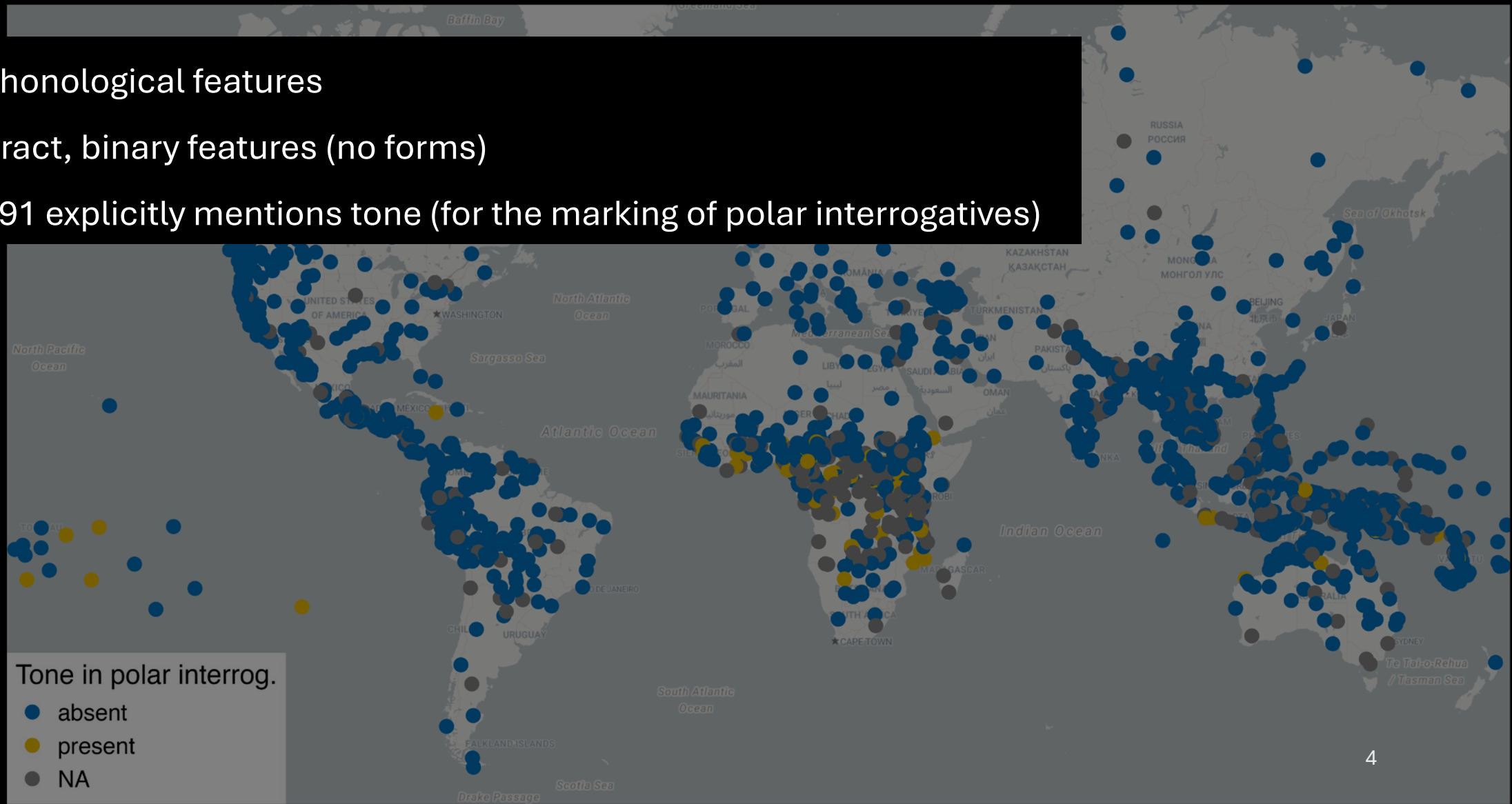
WALS (Dryer & Haspelmath 2013)



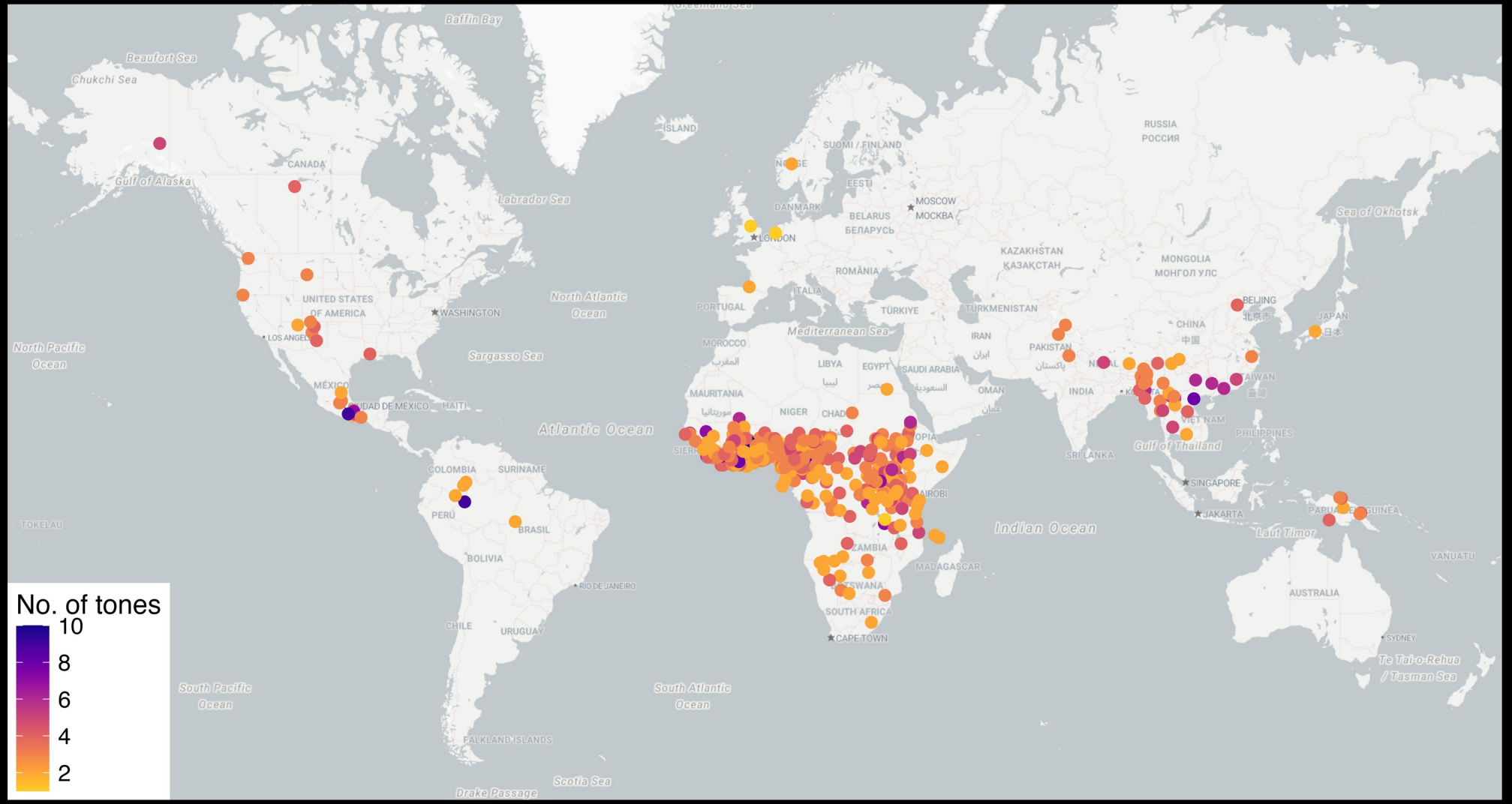
Feature 13A
(Maddieson 2013)

Grambank (Skirgård et al. 2024)

- no phonological features
- abstract, binary features (no forms)
- GB291 explicitly mentions tone (for the marking of polar interrogatives)



PHOIBLE 2.0 (Moran & McCloy 2019)



Area/family/subsystem-specific typologies

Property-driven/canonical tone typology (Hyman 2009, 2015)

tone systems & tone domains in Bodish (Tibeto-Burman) (Hildebrandt 2007)

Contour tone restrictions (Gordon 2001)

Diversity of tone in Papua (Cahill 2011)

Inflectional tone in Otomanguean (Palancar et al. 2016)

Tonal exponence typology (Kaldhol 2024)

Tone and phonation in SEA (Brunelle & Kirby 2016)

Grammatical tone typology (Rolle 2018)

Brunelle & Kirby 2016. LL Compass 10(4) | Cahill 2011. SIL WP | Hildebrandt 2007. New Chall.| Gordon 2001. St. in Lang 23 (3)| Hyman 2009.L Sci 31(2-3); 2015 Tono, accento | Kaldhol 2024. Morph |Palancar et al. Tone & Infl. |

Comparability issues

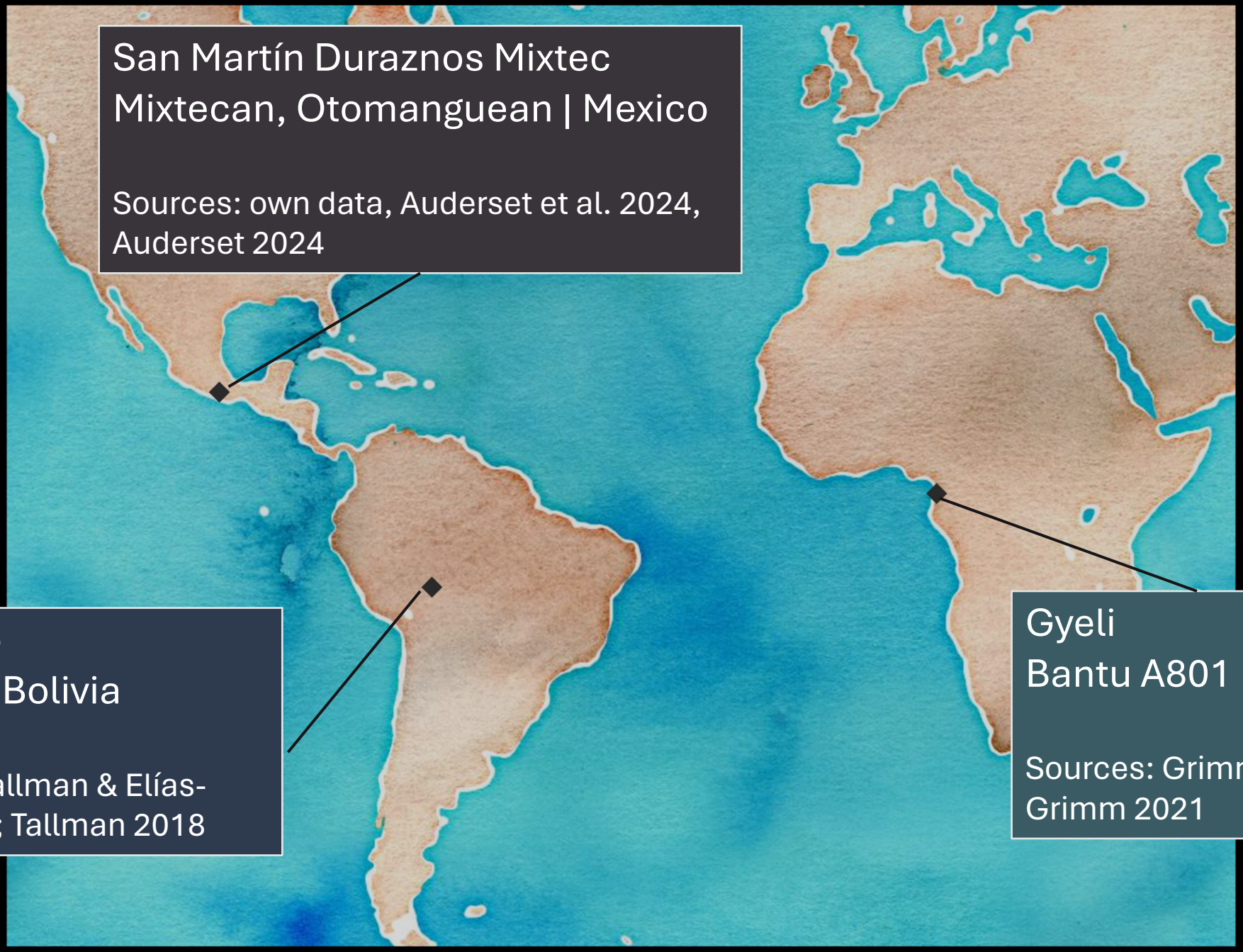
WHAT'S WHERE WHY?
(Bickel 2007)

Comparability issues

"[T]he prosodic category stress in West Germanic languages, which implicitly underlies practically all work on stress, *is a complex cluster concept consisting of at least six dimensions which in turn involve a number of subdimensions. Because of its complexity, this concept is not useful for cross-linguistic comparison*."
(Himmelmann 2023; emphasis mine)

"When we analyse African tonal languages, things get even worse. Instead of a few typology-challenging anomalies as in Tamashek, *almost the whole grammar can become a tissue of language-specific interactions between tonal patterning and grammatical categories*, tenuously held together by subtle language-specific motifs."
(Heath 2016; emphasis mine)

Heath 2016. Ling Typ 20(3) | Himmelmann 2023. Ling Typ 27(2)


San Martín Duraznos Mixtec
Mixtecan, Otomanguean | Mexico
Sources: own data, Auderset et al. 2024, Auderset 2024
Chácobo
Panoan | Bolivia
Sources: Tallman & Elías-Ulloa 2020; Tallman 2018
Gyeli
Bantu A801 | Cameroon
Sources: Grimm 2023; Grimm 2021

Overview of tone inventories and properties

Feature	Duraznos Mixtec	Gyeli	Chácobo
Tones # (with TL vs. without)	H, M, L 3	H, L, R, F, ø 5 ~ 4	H, ø 2 ~ 1
T(one)B(earing)U(nit)	mora	syllable	mora or syllable
Density	every TBU specified for tone	not every TBU specified for tone	not every TBU specified for tone
Culminativity	no	no	yes
Co-occurrence restrictions	no	no	no
Lexical tone	yes	yes	yes
Grammatical tone	yes	yes	yes
Processes	tone sandhi in dependent pronouns, L-tone dissimilation	high tone spreading	floating H-tone, tone sandhi

Gyeli

Phonology

Grammatical tone overwrites lexical tone only when there are not enough toneless syllables

H-tone spreading on toneless TBUs
nominal domain: left to right
verbal domain: right to left

Level	Contour
H	F
L	R
ø (realized as L or H)	

Table 1. *Distribution of level tones in noun stems.*

Tonal pattern	Frequency		Example	
σ	(119/224)	53.1%		
L	57	47.9%	ndɛ̀	'bait'
H	62	52.1%	nká	'line, row'
σσ	(518/555)	93.3%		
L L	115	22.2%	ntɔ̀ngɛ̀	'hornet, wasp'
H H	148	28.6%	njímí	'blind person'
L H	106	20.5%	vìnɔ́	'finger'
H L	150	29.0%	djínɔ̀	'name'
σσσ	(86/90)	95.6%		
L L L	26	29.1%	bɛ̀ngvùdɛ̀	'golden angwantibo'
H H H	14	17.4%	títímɔ́	'middle'
L H H	6	7.0%	ndzìmɔ́zɔ́	'guard'
H L L	13	15.1%	mpí'ìdì	'heat (from fire)'
L H L	10	11.6%	sìsímù	'shadow (of person)'
H L H	3	3.5%	nkúmbɔ̀lɔ́	'diarrhea'
L L H	5	5.8%	mìntùlí	'mouse'
H H L	9	10.5%	dyúngúlɛ̀	'chameleon'

Lexicon

Verbs	Ex.	Gloss
L	bà	PRS.smoke.sth
H	bá	PRS.marry
L ø (ø)	gyìbɔ	PRS. sharpen
H ø (ø)	gyíbɔ	PRS.call

Nouns +	Gloss
síŋgí	squirrel
sìŋgì	spirit
síŋgì	cat
mbɛ̂	door
mbɛ̀	drum

Gyeli

Object-marking H tone: realized on the element immediately following the verb

Grammar/Function

Category	STAMP tone	Verb tone
PRS.F	H	XL
PRS.M	H	XH
INCH.F	LH	XL
INCH.M	LH	XH
PST1	L	XH
PST2	L	XH
FUT	H	XL
IMP	(H)	XHL
SUBJ	H	XH.L

Table 3. *Tense–mood forms with L- and H-toned verb roots.*

TAM	STAMP	σ		σσ		σσσ		STAMP V	
		L	H	L	H	L	H		
	1PL	*bà*	*bá*	*gyìbɔ*	*gyíbɔ*	*pàgala*	*págala*		
PRS_F	yá	bà	bâ	gyìbɔ̀	gyíbɔ̀	pàgàlà	págàlà	H	XL
PRS_M	yá	bá	bá	gyìbɔ́	gyíbɔ́	pàgálá	págálá	H	XH
$INCH_F$	yàá	bà	bâ	gyìbɔ̀	gyíbɔ̀	pàgàlà	págàlà	LH	XL
$INCH_M$	yàá	bá	bá	gyìbɔ́	gyíbɔ́	pàgálá	págálá	LH	XH
$PST1_{F,M}$	yà	bá	bá	gyìbɔ́	gyíbɔ́	pàgálá	págálá	L	XH
$PST2_{F,M}$	yáà	bá	bá	gyìbɔ́	gyíbɔ́	pàgálá	págálá	L	XH
$FUT_{F,M}$	yáà	bà	bâ	gyìbɔ̀	gyíbɔ̀	pàgàlà	págàlà	H	XL
$IMP_{F,M}$	(yá)	bâ	bâ	gyìbɔ̂	gyíbɔ̂	pàgálâ	págálâ	(H)	XHL
$SBJV_{F,M}$	yá	báà	báà	gyìbɔ́ɔ̀	gyíbɔ́ɔ̀	pàgáláà	págáláà	H	XH.L

Gyeli

Phonology

Grammatical tone overwrites lexical tone only when there are not enough toneless syllables

H-tone spreading on toneless TBUs
nominal domain: left to right
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ø (realized as L or H)	

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H ø (ø)	gyíbɔ	PRS.call

Nouns +	Gloss
síŋgí	squirrel
sìŋgì	spirit
síŋgì	cat
mbɛ̂	door
mbɛ̀	drum

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PRS.M	H	XH
INCH.F	LH	XL
INCH.M	LH	XH
PST1	L	XH
PST2	L	XH
FUT	H	XL
IMP	(H)	XHL
SUBJ	H	XH.L

Diachrony

Probable sources of GT:
Segments+Tone > Tone as sole exponent
Lexical tones are reconstructable to Proto-Bantu

Lexicon

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Duraznos Mixtec

Sandhi processes of dependent pronouns conditioning: PoS, tone, and syllable structure of the previous element

Morphosyntax

Toneme	Allotones
H	--
M	H / L_#+L
L	F / M_#+H

Phonology

Melodies (lexical domains)		
H.H	MH	LH
HM	MM	LM
HL	ML	LL

Form	Gloss	PoS
$ni^{1}i^{1}$	salt	N
$ni^{1}i^{3}$	skin	N
$ni^{5}i^{5}$	hail	N
$ka^{3}\tilde{a}^{3}$	make holes	V
$ka^{1}\tilde{a}^{3}$	get used to	V

Lexicon/ Contrasts

Duraznos M.

POT	INCPL	COMPL	Gloss	Class
ɕa^{1}a^{1}	ɕa^{5}a^{1}	i^{1}ɕa^{1}a^{1}	‘arrive’	1a
ka^{3}sũ^{3}	ka^{5}sũ^{5}	i^{1}ka^{3}sũ^{3}	‘be fried’	1b
ku^{3}tʃi^{3}	ʃi^{5}tʃi^{3}	i^{1}ʃi^{3}tʃi^{3}	‘take a bath’	2a
ka^{1}ʔnu^{1}	ɕa^{5}ʔnu^{1}	i^{1}ɕa^{1}ʔnu^{1}	‘break, fold’	3a
ka^{3}tɕa^{3}	ɕa^{5}tɕa^{5}	i^{1}ɕa^{3}tɕa^{3}	‘throw, dig up’	3b
ku^{3}na^{1}ʔa^{1}	na^{5}ʔa^{1}	i^{1}ɕi^{1}na^{1}ʔa^{1}	‘look like’	4a
ku^{3}ʒa^{5}ʔa^{5}	ʒa^{5}ʔa^{5}	i^{1}ku^{3}ʒa^{5}ʔa^{5}	‘be yellow’	4b

Grammar/Function

Derivational H:
N → Vstat
(not productive)

Class	POT			INCPL		COMPL		
Morae	Prfx	μ1	μ2	μ1	μ2	Prfx	μ1	μ1
1a		X	X	H	X	i^{1}	X	X
1b		X	X	H	H	i^{1}	X	X
2		kX	X	ɕH	X	i^{1}	ɕX	X
3	ku^{3}	X	X	H	X	i^{1}-ɕi^{1}	X	X
4	ku^{3}	X	X	H	X	i^{1}-ku^{3}	X	X

Duraznos Mixtec

Sandhi processes of dependent pronouns conditioning: PoS, tone, and syllable structure of the previous element

Phonology

Toneme	Allotones
H (5)	--
M (3)	H / L_#+L
L (1)	F / M_#+H

Melodies (lexical domains		
HH	MH	LH
HM	MM	LM
HL	ML	LL

Grammar/Function

Derivational H:
N → Vstat
(not productive)

Class	POT			INCPL		COMPL		
Morae	Prfx	μ1	μ2	μ1	μ2	Prfx	μ1	μ1
1a		X	X	H	X	i^1	X	X
1b		X	X	H	H	i^1	X	X
2		kX	X	ɕH	X	i^1	ɕX	X
3	ku^3	X	X	H	X	i^1-ɕi^1	X	X
4	ku^3	X	X	H	X	i^1-ku^3	X	X

Lexicon/ Contrasts

Form	Gloss	PoS
ni^1i^1	salt	N
ni^1i^3	skin	N
ni^5i^5	hail	N
$ka^3\tilde{a}^3$	make holes	V
$ka^1\tilde{a}^3$	get used to	V

Duraznos Mixtec

Sandhi processes of dependent pronouns conditioning: PoS, tone, and syllable structure of the previous element

Phonology

Toneme	Allotones
H	--
M	H / L_#+L
L	F / M_#+H

Melodies (lexical domains		
HH	MH	LH
HM	MM	LM
HL	ML	LL

Proto-Mixtec

*H > M	*?.L > H
*L > L	*?.H > H
*Hʔ# > L	INCPL: *H > H
*Lʔ# > L	

Diachrony

Grammar/Function

Derivational H:
N → Vstat
(not productive)

Class	POT			INCPL		COMPL		
Morae	Prfx	μ1	μ2	μ1	μ2	Prfx	μ1	μ1
1a		X	X	H	X	i^1	X	X
1b		X	X	H	H	i^1	X	X
2		kX	X	ɕH	X	i^1	ɕX	X
3	ku^3	X	X	H	X	i^1-ɕi^1	X	X
4	ku^3	X	X	H	X	i^1-ku^3	X	X

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Lexicon/
Contrasts

Chácobo

Phonetics

Category	Primary corr.	Secondary corr.
High Tone	pitch	duration
Stress	intensity	duration

Phonology

Category	Domain	Realization
High Tone	any syll.	H ~ L
Toneless	any syll.	L ~ 0
Stress	first two syll.	#H ~ #M

Lexicon /Contrasts

Form	Gloss	PoS
káʂa	be angry	V intr.
kaʂá	play	V intr.
áni	lake	N
aní	grow	V
raβi	Rabi	N
rámi	Rami	

Chácobo

Grammar/Function

ERGATIVE/GENITIVE:
H on final syllable of noun phrase erases all other H on the element it appears on

(24) [tʃá.ʂo **ka.ma.nó** kɨ.jó.ɾo.ʔá.kɨ]
tʃáʂo kamáno=´ kɨjó=ɾoʔá= kɨ
deer jaguar=erg finish=limit=dec:pst
"The jaguar only finished the deer"

Lexicon /Contrasts

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kaʂá	play	V intr.
áni	lake	N
aní	grow	V
raβi	Rabi	N
rámi	Rami	

Morphosyntax

Process	Target	Constituency
Min. tone insertion	Prefixes to Adverbials	do not converge with other phonological or morphosyntactic constituents
Max. tone insertion	up to clause type	
Min. tone deletion	NP to end of clause	
Max. tone deletion	whole clause	

Chácobo

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Morphosyntax

Process	Target	Constituency
Min. tone insertion	Prefixes to Adverbials (7-17)	do not converge with other phonological or morphosyntactic constituents
Max. tone insertion	up to clause type (1-24)	
Min. tone deletion	NP to end of clause (6-27)	
Max. tone deletion	whole clause (1-27)	

Chácobo

Diachrony

Source of H:
stressed heavy syllables

Phonetics

Category	Primary corr.	Secondary corr.
High Tone	pitch	duration
Stress	intensity	duration

Grammar/Functi

Probable source of ERG:
suffix –n + stress shift

ERGATIVE/GENITIVE:
H on final syllable of noun phrase erases all other H on the element it appears on

Phonology

Category	Domain	Realization
High Tone	any syll.	H ~ L
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Stress	first two syll.	#H ~ #M

Lexicon /Contrasts

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Max. tone insertion	up to clause type (1-24)	
Min. tone deletion	NP to end of clause (6-27)	
Max. tone deletion	whole clause (1-27)	

Morphosyntax

A synthesis

Phonetics
- phonetic correlates
- phonetic variation (predictability ~ allotones, etc.)

Sociolinguistics
- multilingualism
- tones as identity markers
- word play/creativity

Phonology
- tones (tonal primitives)
- TBU
- distributional properties (culminativity, privativity, sandhi, etc.)

Diachrony
- reconstruction and proto-tones
- correspondences and pathways
- sources

Lexicon
- contrasts
- domains/melodies
- word class sensitivity

Grammar and Function:
- sole or co-exponent of grammatical categories
- as a part of inflectional class systems
- tonal processes sensitive to morphosyntax

Conclusions and outlook

- tone evades easy classification both into coarse-grained types and into more fine-grained, abstract, codable traits
- tones forces us see to see linguistic systems as interconnected
- diachrony is an essential component for explanation

- tone is not the problem – it just shows us where we need to go

classification into broad, language-level types

multivariate/ distributional typology

family-level diachronic (micro)typology & comparison of evolutionary processes

Conclusions and outlook

- collaboration with fieldworkers & language experts
- corpora, naturalistic speech
- towards a true historicization, focusing not on synchronic distributions of traits but on the evolution of systems

EVOTONE
PI: James Kirby
https://evotone.github.io/
origins, acquisition, evolution of tone

Qualitative and quantitative approaches to tone
(Auderset 2024. L Dyn & Change 14 (10))

Theory of tone
PI: Valentin Vydrin
https://thot.huma-num.fr/
role of tone in phon. systems

Integration of tone into constituency typology
(Tallman et al. 2024. LangSciP)

classification into broad, language-level types

multivariate/ distributional typology

family-level diachronic (micro)typology & comparison of evolutionary processes

Acknowledgements

I would like to thank (in alphabetical order):

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Naomi Peck
Steven Moran
Uta Reinöhl

for discussion and comments.

"I never met a tone I didn't like."
(Hyman 2018)

Tone in Gyeli (Bantu A810; Cameroon)

- Contrastive tones (tonemes): 5 (or 4 if toneless is not counted)
  - high, low, rising, falling, toneless
- TBU: syllable
- lexical tone: yes
- grammatical tone: yes
- density: not every TBU is specified for tone
  - nouns are generally specified for tones, verbs only on the first syllable, other PoS are often toneless
- co-occurrence restrictions: none
- culminativity: no
- processes: high tone spreading

Sources: Grimm 2023; Grimm 2021

Tones in Gyeli

Table 5. *Distribution of underlying tones in verb stems.*

Tonal pattern		Frequency		Example	
σ		(88)			
	L	39	44.3%	*kɛ̀*	'go'
	H [HL]	49	55.7%	*nyɛ̂*	'see'
σσ		(213)			
	L ∅	92	45.2%	*sèngɛ*	'lower'
	H ∅	121	56.8%	*gyíbɔ*	'call'
σσσ		(76)			
	L ∅ ∅	26	34.2%	*kàsɛlɛ*	'light'
	H ∅ ∅	50	65.8%	*djímɛsɛ*	'extinguish'

Tone in Chácobo (Pano; Bolivia)

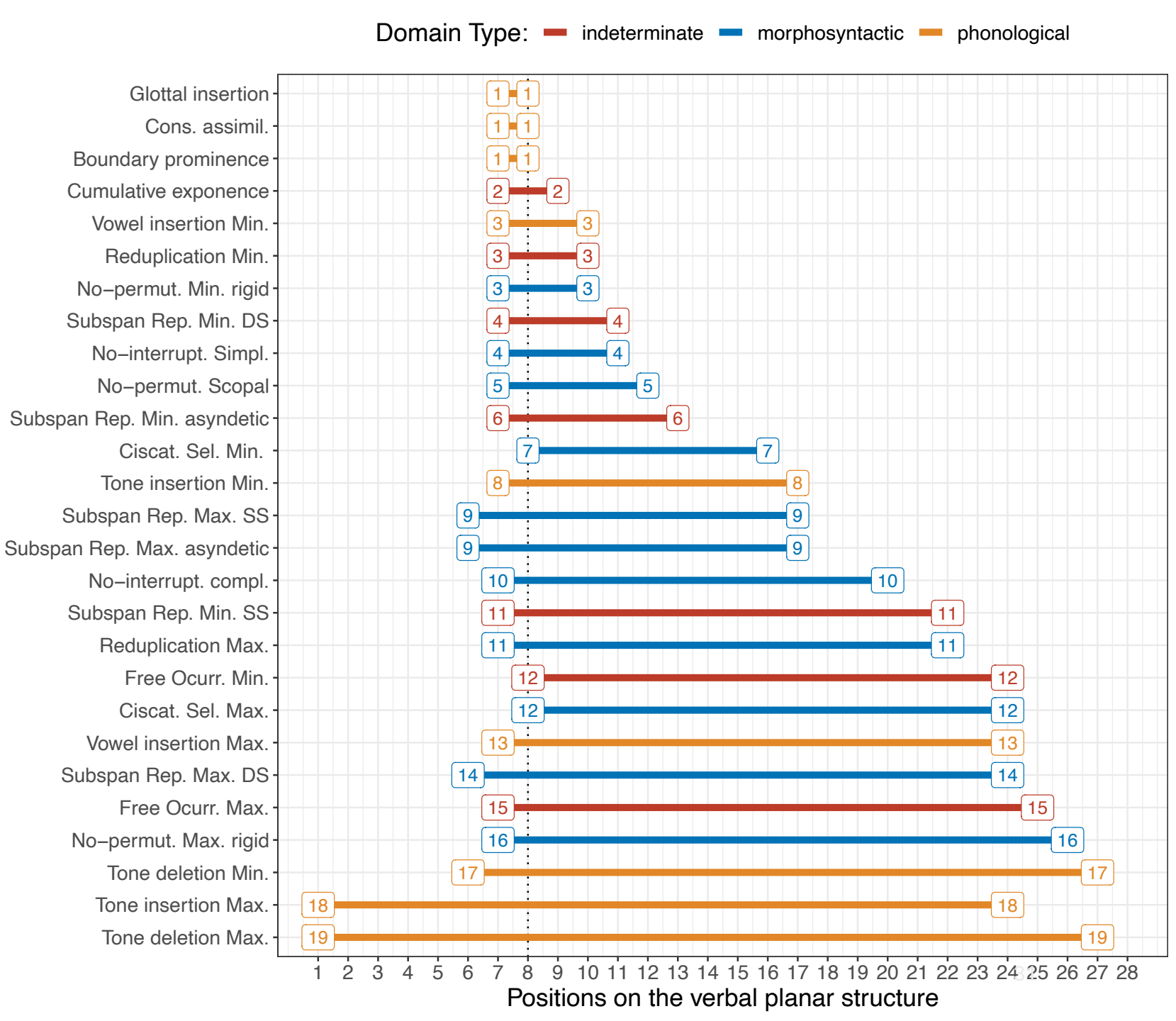
- Contrastive tones (tonemes): 2 (or 1 if toneless is not counted)
  - high, toneless
- TBU: syllable
- lexical tone: yes
- grammatical tone: yes
- density: not every TBU is specified for tone
  - roots can be specified for tone or not
- co-occurrence restrictions: certain H block other H from surfacing
- culminativity: yes, one H-tone per morpheme
- processes: floating H-tone, tone sandhi

Sources: Tallman & Elías-Ulloa 2020; Tallman 2018

(9) [ɨ.a. tsá.ja.kɨ. ɾa.ßi̯]
/ɨ-*a* *tsaja=kɨ* ***rabi/***
1sg-acc see=dec:past **Rabi**
“Rabi has looked at me.”

(10) [ɨ.a. tsá.ja.kɨ. ɾá.mi]
/ɨ-*a* *tsaja=kɨ* ɾámi/
1sg-acc see=dec:past **Rami**
“Rami has looked at me.”

Chácobo Constituency



Tone in Duraznos Mixtec (Mixtecan; Mexico)

- Contrastive tones (tonemes): 3
  - high, mid, low
- TBU: mora
- lexical tone: yes
- grammatical tone: yes
- density: every TBU is specified for tone
- co-occurrence restrictions: no
- culminativity: no
- processes: tone sandhi in pronouns, L-tone dissimilation

Sources: Auderset et al. 2024; own data and analysis

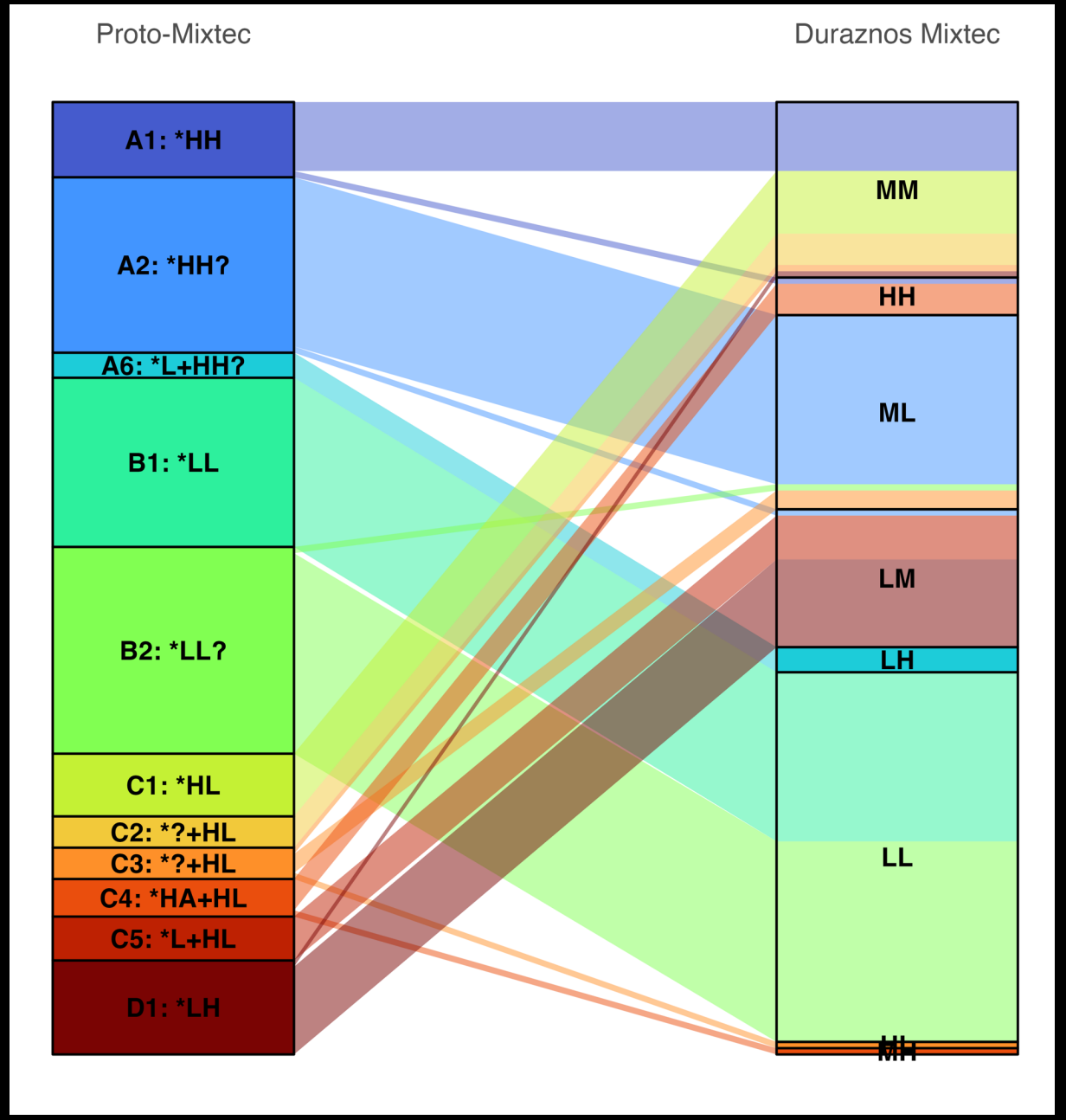
pronombre	sustantivos	verbos
1SG ì	XB + i^1 > AB	
1PL.INCL ò	[AB]ʔ(C)M + o^1 > o^5 **MʔM + ɾa^1 > ɾa^3**	[AB]ʔ(C)M + o^1 > o^5 **M(ʔ)M + ɾa^1 > ɾa^3**
1PL.EXCL ntì	[AB]ʔM + ^{n}di^1 > ^{n}di^3 [MB](ʔ)B + ^{n}di^1 > ^{n}di^3 **MʔM + ^{n}di^1 > ^{n}di^3**	[AB]ʔM + ^{n}di^1 > ^{n}di^3 [MB](ʔ)B + ^{n}di^1 > ^{n}di^3 **M(ʔ)M + ^{n}di^1 > ^{n}di^3**
2SG.INF un	copia el tono anterior XʔM + ũ > ũ^5	
2PL ntò	[AB]ʔ(C)M + ^{n}do^1 > ^{n}do^5 **MʔM + ^{n}do^1 > ^{n}do^5**	[AB]ʔ(C)M + ^{n}do^1 > ^{n}do^5 **M(ʔ)M + ^{n}do^1 > ^{n}do^5**
3SG.M rà	[AB]ʔM + ɾa^1 > ra^3 [MB](ʔ)B + ɾa^1 > ra^3 **MʔM + ɾa^1 > ɾa^3**	[AB]ʔM + ɾa^1 > ɾa^3 [MB](ʔ)B + ɾa^1 > ɾa^3 **M(ʔ)M + ɾa^1 > ɾa^3**
3SG.F àn ~ ña	[AB]ʔ(C)M + ã^1/ɲa^1 > ã^5/ɲa^5 **MʔM** + ã^1/ɲa^1 > ã^5/ɲa^5	[AB]ʔ(C)M + ã^1/ɲa^1 > ã^5/ɲa^5 **M(ʔ)M** + ã^1/ɲa^1 > ã^5/ɲa^5
3PL nà	[AB]ʔ(C)M + na^1 > na^5 **M(ʔ)M + na^1 > na^5**	[AB]ʔ(C)M + na^1 > na^5 **M(ʔ)M + na^1 > na^5**

Inflectional classes in San Martín Duraznos Mixtec

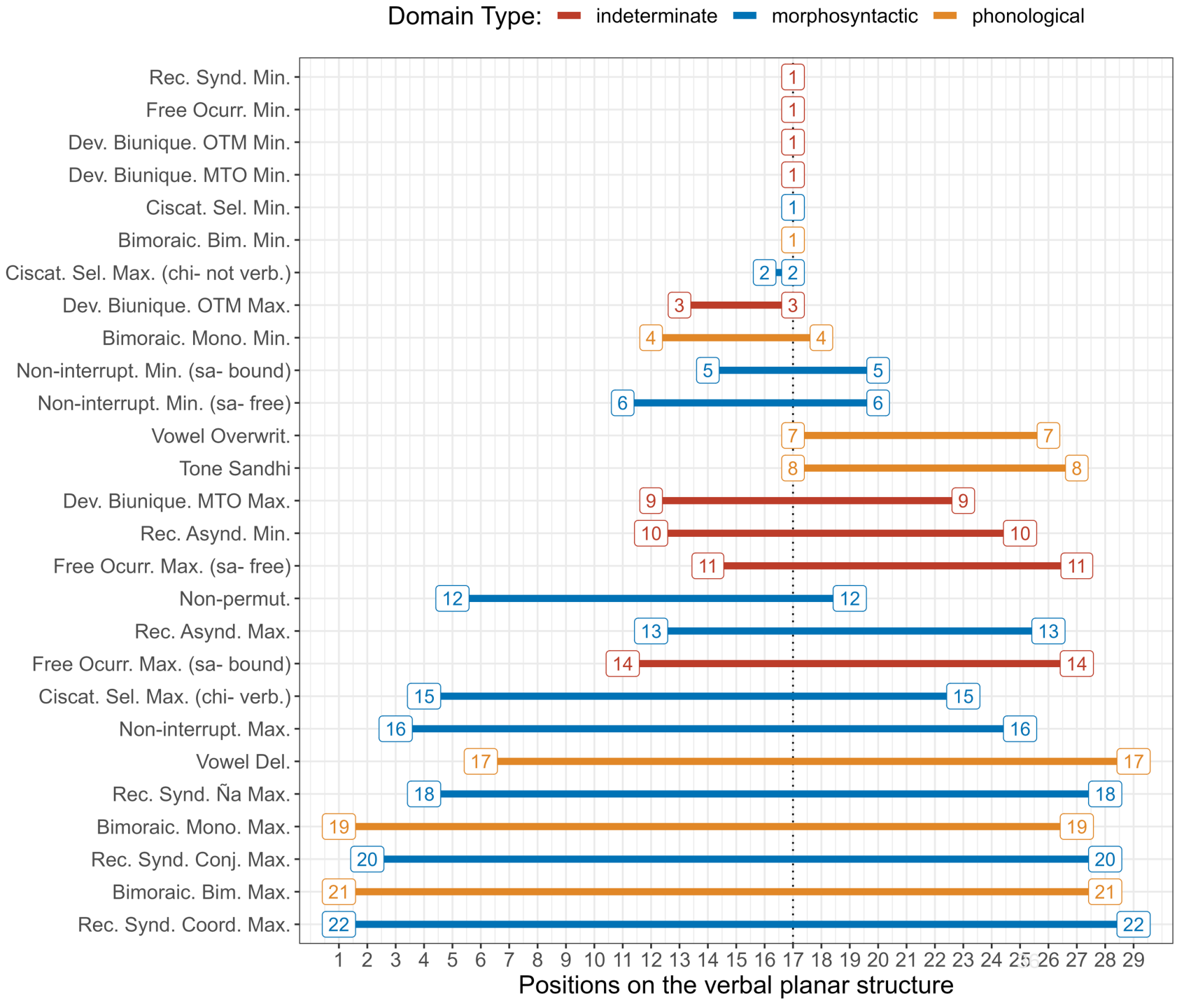
POT	INCPL	COMPL	Gloss	Class	B&M 1982
ɕa^{1}a^{1}	ɕa^{5}a^{1}	i^{1}ɕa^{1}a^{1}	‘arrive’	1a	C / reg.
ka^{3}sũ^{3}	ka^{5}sũ^{5}	i^{1}ka^{3}sũ^{3}	‘be fried’	1b	C / reg.
ku^{3}tʃi^{3}	ʃi^{5}tʃi^{3}	i^{1}ʃi^{3}tʃi^{3}	‘take a bath’	2a	C / irreg.
ka^{1}ʔnu^{1}	ɕa^{5}ʔnu^{1}	i^{1}ɕa^{1}ʔnu^{1}	‘break, fold’	3a	V
ka^{3}tɕa^{3}	ɕa^{5}tɕa^{5}	i^{1}ɕa^{3}tɕa^{3}	‘throw, dig up’	3b	V
ku^{3}na^{1}ʔa^{1}	na^{5}ʔa^{1}	i^{1}ɕi^{1}na^{1}ʔa^{1}	‘look like’	4a	C
ku^{3}ʒa^{5}ʔa^{5}	ʒa^{5}ʔa^{5}	i^{1}ku^{3}ʒa^{5}ʔa^{5}	‘be yellow’	4b	C
ko^{3}o^{3}	i$^{5\sim3}$	i^{1}ɕo^{1}o^{3}	‘live, stay’	5	irregular
ku^{3}ni^{3}	ɕi^{3}tu^{3}ntʑe^{3}ʔe^{3}	i^{1}ɕi^{1}ni^{3}	‘see sth.’	5	irregular

Main classes of simplex verbs

Mixtec tone diachrony



Duraznos Mixtec Constituency



Tone in Peras Mixtec (Mixtecan; Mexico)

- Contrastive tones (tonemes): 5
  - high, mid, low, rising, falling
- TBU: mora
- lexical tone: yes
- grammatical tone: yes
- density: every TBU is specified for tone
- co-occurrence restrictions: contours restricted to final TBU
- culminativity: no
- processes: tone sandhi in pronouns

Sources: Auderset et al. 2024; own data and analysis

New approaches in the comparison of tone

EVOTONE: The emergence and evolution of linguistic tone

The EVOTONE project studies the **origins, acquisition, and evolution** of linguistic tone: the use of pitch to distinguish between the meaning of words. … EVOTONE will provide the first comprehensive study of tonal emergence and evolution, combining detailed **phonetic and perceptual studies** with innovative **experimental methodologies** and large-scale computational analysis of the structural principles correlated with the emergence of tone.

PI: James Kirby
https://evotone.github.io/

ThoT: Theory of Tone

The aim of the project is to systematically describe the **diversity of the tonal systems** of the languages of the world. The project will accumulate comprehensive data on the **role of tones in phonological systems** of the world's languages: structure of tonal systems, functional load of tone (expressed in the terms of tonal density) in each particular language system, role of tone in **distinguishing lexical and/or grammatical meanings**.

PI: Valentin Vydrin
https://thot.huma-num.fr/

New comparative approaches integrating tone

Constituency in cross-linguistic perspective

… is a consortium, primarily composed of **field researchers**, who seek to understand the **notion of morphological, syntactic and phonological constituency** across the languages of the world. This is done by attempting to dissect the notion of 'constituency test', to develop an extensible set of variables for cross-linguistic comparison. … The overall goal is to develop to understand **variation in linguistic architecture** (i.e. the division of language structure into more or less distinct modules).

PI: Adam J. R. Tallman
https://sites.google.com/view/constituencytypology

Diachronic comparative approaches

Dockum 2019: shows that the Comparative Method can be applied to tone; based on extensive data from Kra-Dai

Auderset 2024: shows that rates of change in tones are not faster or less informative for subgrouping than segments; based on extensive data from Mixtec